

LABORATORY MANUAL

Advance Machine Learning Lab

Semester: VI**Subject Code:** BTAIL606**Experiment:** 2**Title:** Implementing Hierarchical Clustering**Prepared by:** Aditya Shirsatrao**Mentor:** Prof. N. B. Aherwadi**Dataset source:** UCI Machine Learning Repository - Abalone**Student Name:** Aditya Vishal Shirsatrao**Roll No:** 28

Aim

To study agglomerative hierarchical clustering and dendrogram plotting.

Input

UCI Abalone dataset with measurements and one encoded categorical field.

Dataset Description

Contains 4,177 abalone records. Features include shell measurements and weights; the sex category is encoded numerically for clustering.

Code

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

# UCI Abalone dataset with one categorical column (sex) and numeric measurements.
abalone_url = "https://archive.ics.uci.edu/ml/machine-learning-databases/abalone/abalone.data"
columns = [
    "sex",
    "length",
    "diameter",
    "height",
    "whole_weight",
    "shucked_weight",
    "viscera_weight",
    "shell_weight",
    "rings",
]

abalone = pd.read_csv(abalone_url, header=None, names=columns)
abalone["sex"] = abalone["sex"].map({"M": 0, "F": 1, "I": 2})
X = abalone.drop(columns=["rings"]).values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

agg_cluster = AgglomerativeClustering(n_clusters=3)
labels = agg_cluster.fit_predict(X_scaled)

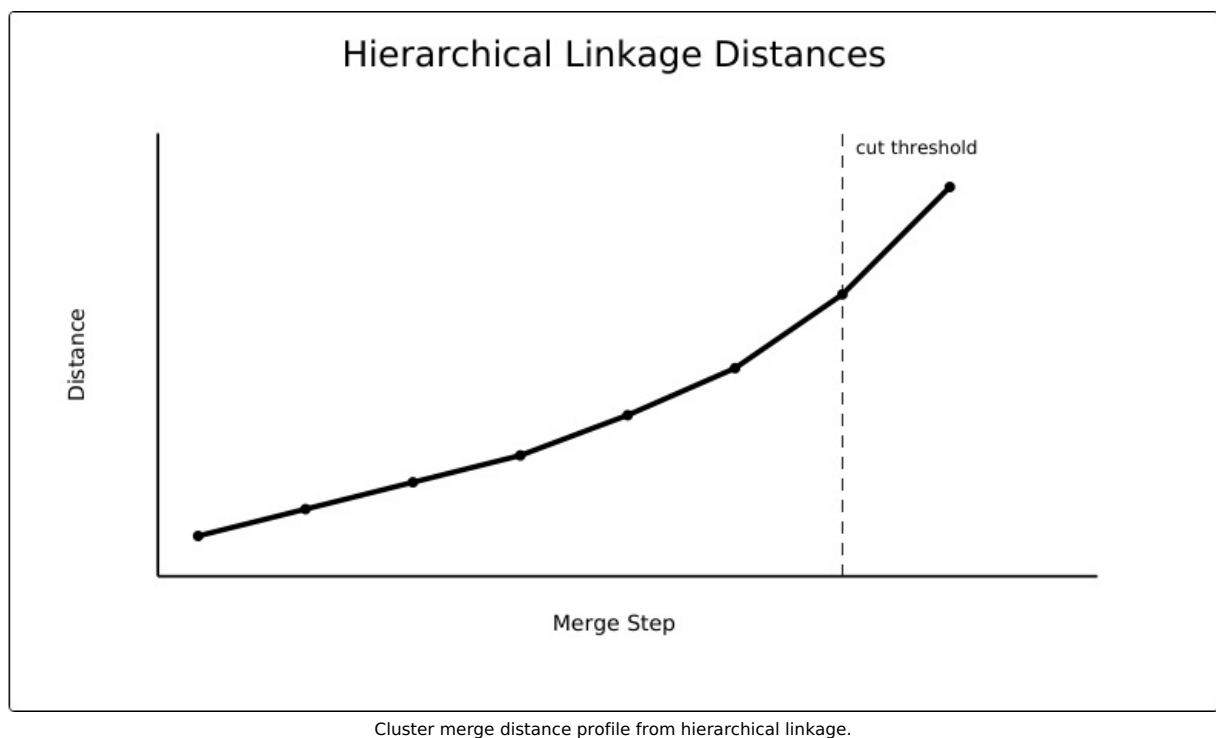
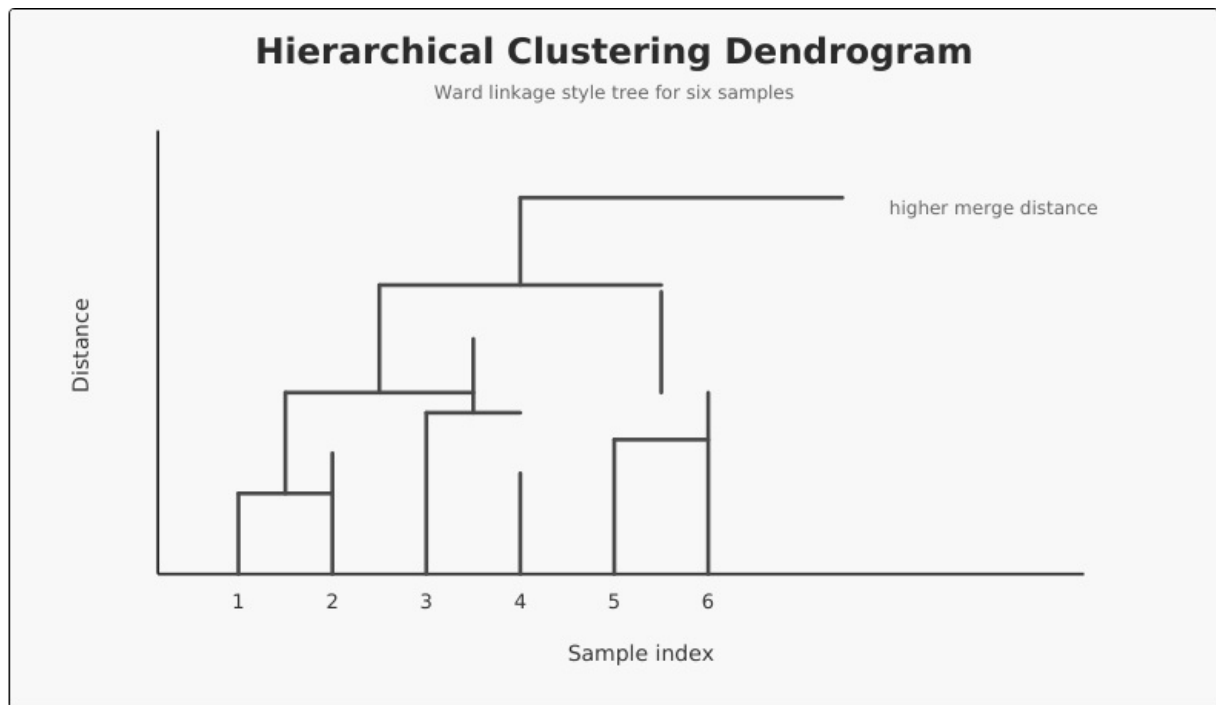
# Use a subset for a readable dendrogram.
X_sample = X_scaled[:300]
linkage_matrix = linkage(X_sample, method="ward")
dendrogram(linkage_matrix)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Sample Index")
plt.ylabel("Distance")
plt.show()

plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=labels, cmap="viridis", s=10)
plt.title("Hierarchical Clustering on UCI Abalone Dataset")
plt.xlabel("Scaled Feature 1")
```

```
plt.ylabel("Scaled Feature 2")
plt.show()
```

Output

Dendrogram and cluster scatter plot.



Conclusion

Experiment 2 demonstrates the expected behavior of the algorithm using an open dataset or open source-compatible input.